Problem Chosen	2024 HiMCM/MidMCM Summary Sheet	Team Control Number
A		14995

Research on an Olympic SDEs Selection Model Based on Combine Weighting Method (CWM)

Summary

To ensure the modernity and global appeal of Olympic events, thereby catering to the demands of contemporary audiences and achieving broader cultural inclusivity [1], a **comprehensive evaluation and event selection model** has been established based on the standards of the International Olympic Committee (IOC). This model takes into account factors such as popularity, gender equality, sustainability, safety and fair competition, as well as economic benefits, aiming to provide a scientific basis for the IOC's decision-making regarding new sports, disciplines, and events (SDEs).

Task 1. Based on the standards of the IOC, we have selected seven evaluation indicators: **popularity and accessibility**, gender equality, sustainability, **inclusivity**, **relevance and innovation**, **safety and fair competition**, as well as **economic benefits and commercialization potential**. Quantitative calculation expressions have been established for each indicator, and the values of these seven indicators for 48 Olympic events, including aquatic sports, archery, athletics, ice hockey, etc., have been calculated. The detailed results are shown in Table 3.

Task 2. We established a **Combined Weighting Method (CWM)** and **MC-TOPSIS (Monte Carlo Simulation TOPSIS) Model** to determine the SDEs that best meet the requirements of the IOC. Firstly, the Analytic Hierarchy Process (AHP), Entropy Weight Method, and Coefficient of Variation Method were combined to calculate the weight matrix. Then, the CRITIC method was used to determine the final indicator weight coefficients, reflecting the degree of influence of different factors on the selection of SDEs. Based on this, the score rankings of the 48 events were determined, with **the top three being Athletics, Football, and Basketball**. Subsequently, through comparison using MC-TOPSIS (Monte Carlo Simulation TOPSIS), which provides multiple simulations to ensure the robustness and anti-volatility of the scores, the SDEs that best meet the requirements of the IOC were identified. **The top three then changed to Football, Athletics, and Basketball**.

Task 3. We selected three SDEs (new or removed from the Olympics): Karate, Sport Climbing, and Offshore Rowing. Additionally, we chose six events from the existing major categories: Track and Field, Shooting, Basketball, Gymnastics, Boxing, and Aquatic Sports. According to the model's calculations, the scores were as follows: **Karate scored 22.674, Sport Climbing scored 33.268, and Offshore Rowing scored 9.4959. The scores revealed varying performances among these three newly added events. The remaining six events are traditional Olympic sports with a stable audience base and global participation**. By evaluating these sports, we verified the efficiency and applicability of the model across different events.

Task 4. Using the simulation model we developed, we recommend **Cricket, Martial Arts, and Squash** as priority new events for the 2032 Olympics. Data indicates that Cricket boasts a significant global audience base and economic benefits, Martial Arts performs exceptionally well in terms of cultural diversity and gender equality, and Squash aligns with sustainable development principles. Additionally, we have identified **potential events for 2036 and beyond: Esports and Ultimate Frisbee**, to cater to the interests of the new generation of audiences.

Task 5. We employed multiple weighting methods and Monte Carlo simulations to ensure the robustness of our model. This model provides multidimensional support for the selection of SDEs and offers a scientific basis for the IOC to make balanced decisions between tradition and innovation.

Finally, we provided the IOC with scientific decision-making evidence and conclusions for the selection of SDEs.

Keywords: Combine Weighting Method, MC-TOPSIS, IOC, SDEs, Evaluation Factors

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1 INTRODUCTION

1.1 Background

The International Olympic Committee (IOC) is preparing for the 2032 Summer Olympics in Brisbane, with plans to adjust the Olympic events in accordance with contemporary trends in global sports and the demands of the modern era. Throughout the history of the Olympic Games, there have been instances where certain sports and events (SDEs) have been introduced, removed, or even reintroduced into the Olympic program. In order to guarantee the continued relevance and appeal of the Olympic Games, it is essential that the International Olympic Committee (IOC) conducts a thorough assessment of the sports and events (SDEs) that should be included or removed from the programme. However, in the context of the real world, a multitude of factors influence the potential selection of an event by the International Olympic Committee for inclusion in the Olympic Games. Such factors include, but are not limited to, the number of viewers, the gender ratio, the discrepancy in prize money, the number of participating countries, and the number of doping tests.

1.2 Problem Restatment

The object for the problem is building one or more mathematical models to help the IOC evaluate which sports are best suited for the 2032 Olympics development vision, providing recommendations for adding or removing SDEs, and testing the modelÕs robustness. The evaluation criteria in problems includes popularity, gender equality, sustainability, accessibility, innovation, safety and fair competition. All criteria can be quantitative or qualitative, constant or variable, deterministic or probabilistic. At the same time, the problem also asked to do following task:

1. Select at least three new or removed sports from the 2020, 2024, and 2028 Olympic Games for evaluation.
2. Select at least three projects for evaluation that have been in continuous operation since 1988 or later.
3. Provide priority recommendations for adding or returning projects in 2032.
4. Assess whether there are SDEs with the potential to be included in the 2036 Olympics or later.
5. Assess the robustness of the model and analyze key factors in the model and their impact on recommended results.

Besides, a one-to-two-page summary letter is required for describing the theoretical rationale for the model, demonstrating the model's support for the results of the SDEs assessment, and including a robustness analysis.

1.3 Our Work

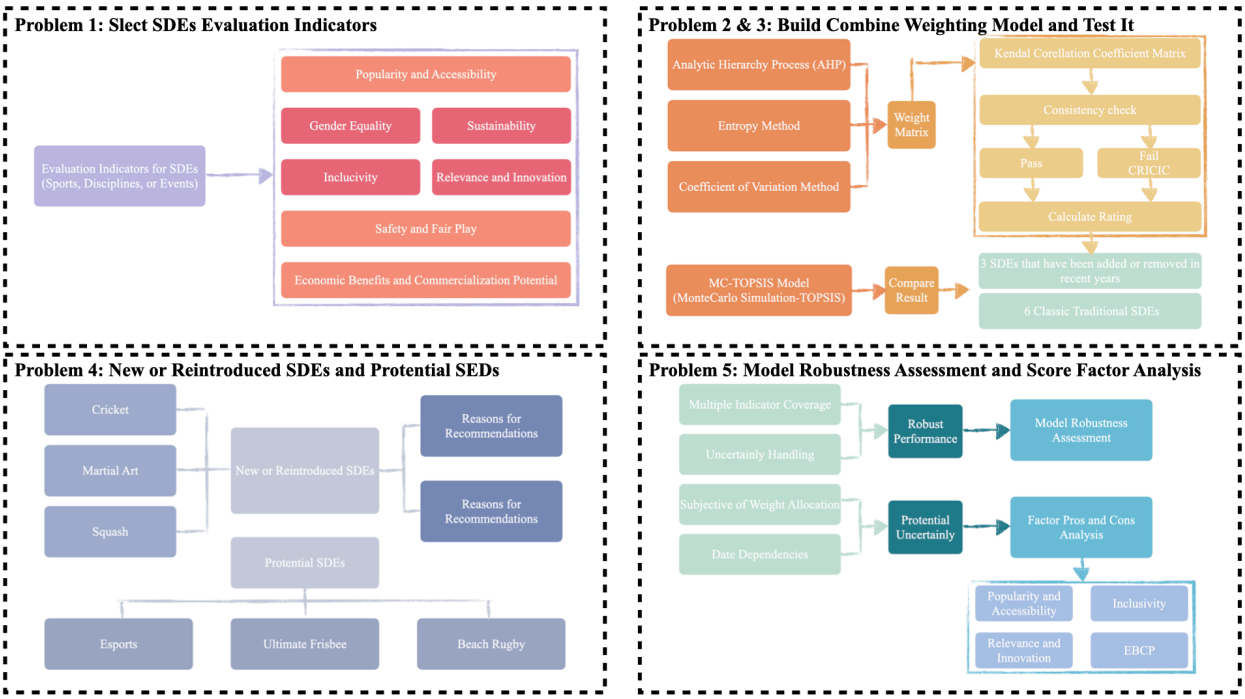


Figure 1: Model Construction

2 Model Assumption

To construct an evaluation model for analyzing the potential new or reintroduced sports disciplines (SDEs) at the 2032 Brisbane Olympic Games, the following assumptions will be utilized. Assumptions in models are listed in Table 1.

Table 1: Assumptions for Model

#	Assumption	Justification
1	Popularity and accessibility are important factors for evaluating programme	Popularity and Accessibility influence the global attention and audience reach of the Olympic Games, making them two key factors for evaluating programs [2]. The breadth of the global audience's number and distribution measures a program's market potential. For instance, cricket boasts a global audience of 2.5 billion, with popularity in Australia, South Asia, and Commonwealth countries as key factors.
2	International Promotion and Inclusiveness are Core Values for Olympic Selection	The Olympic spirit emphasizes diversity and inclusiveness. The degree of internationalization of a sport (e.g., the number of participating countries) reflects its inclusiveness and promotion potential. Sports such as martial arts and ultimate frisbee have widespread participation across multiple countries, aligning with the Olympic goals of globalization and diversity.

Continued on next page

#	Assumption	Justification
3	Economic Benefits and Commercialization Potential are Crucial for Gender Equity of the Programmes	The commercial appeal of Olympic events can generate sponsorship and media revenue, which is pivotal in supporting the long-term development of the Olympic Games. Events with high economic market value are more attractive and sustainable; for instance, cricket's global market value of USD 6.2 billion is one of the reasons for its consideration as a prime candidate.
4	Emerging Sports Should be Innovative and Able to Attract Younger Audiences	The Olympics need to introduce sports that appeal to younger audiences to maintain vitality and modernity. Sports that attract younger audiences align with current Olympic trends, such as esports and ultimate frisbee, which have large young viewer bases and are gradually gaining popularity.
5	Sustainability and Environmental Impact are Important Considerations	The Olympics need to introduce sports that appeal to younger audiences to maintain vitality and modernity. Sports that attract younger audiences align with current Olympic trends, such as esports and ultimate frisbee, which have large young viewer bases and are gradually gaining popularity [3].
6	Traditional Sports Must Maintain Stability and a Strong Audience Base	While the Olympics emphasize innovation and diversity, the stability and audience base of traditional sports remain important. Events like athletics and swimming should be retained in the list to ensure the Olympics balance historical significance with modern trends.

3 Symbols and Notations

Table presents the symbols utilized in the subsequent essay, accompanied by their respective definitions.

Table 2: Glossary of Terms for Evaluation Methodology

Symbol	Definition
n	Number of factors.
a_{ij}	The elements in the judgment matrix A represent the relative importance ratio between the i-th factor and the j-th factor.
p_{ij}	The weight of the i-th evaluation object on the j-th indicator.
τ	Kendall correlation coefficient between the i-th weight vector and the j-th weight vector.
W_i	Weight vector generated by different weighting methods.
A	Data matrix, containing the data of all evaluation objects on various indicators.
S	Comprehensive score.

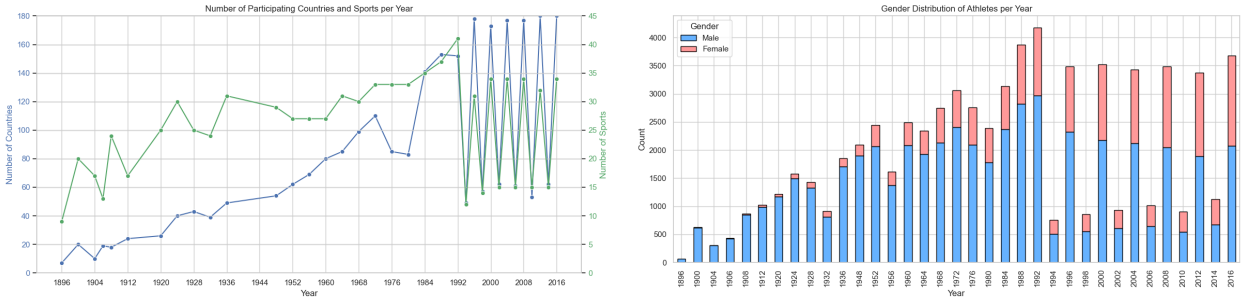
4 Calculation for Evaluation Indicators

IOC selected seven evaluation indicators, such as popularity, gender equality and sustainability, as the fundamental criteria for determining whether a sport should be included in the Olympic Games. While quantitative data is essential for evaluating these indicators, qualitative analysis is also crucial. To make sure the comprehensiveness of the evaluation, it is essential to categorize the nature of the evaluation indicators

in detail.

4.1 Preparing for Calculating Evaluation Indicators

According to data, some of which are given in Figure 2, and IOC criteria, there are several important charasterictics that can be regarded as evaluation indicators, which are listed below.



(a) Age Distribution of Athletes in Each Olympic Games

(b) Gender Statistics of Athletes in Each Olympic Games

Figure 2: Some Statistical Data for Past Olympic Games

4.1.1 Popularity and Accessibility

This criterion reflects the global popularity of a particular sport and the degree of convenience associated with participation, which is the core of the attractiveness of the Olympic Games. Several factors, such as global viewership (television, social media interaction), participation, and geographic coverage, are involved in measuring the criteria. The indicator can be calculated using the following formula:

$$S_{\text{pop}} = \alpha_1 \cdot \frac{P_{\text{g}}}{P_{\text{max}}} + \alpha_2 \cdot \frac{N_{\text{c}}}{N_{\text{max}}} + \alpha_3 \cdot \frac{R_{\text{b}}}{R_{\text{max}}} \tag{1}$$

where P_{g} is global audience size (e.g., television and online viewers); N_{c} is number of countries participating in the sport; R_{b} is media exposure (e.g., number of reports per year on the sport); N_{max} is maximum number of participating countries among all candidate events; P_{max} is maximum global audience size among all candidate events; R_{max} is maximum media exposure among all candidate events; and α_1, α_2, and α_3 are weight coefficients, usually assigned based on the importance of each indicator (e.g., $\alpha_1 = 0.4$, $\alpha_2 = 0.4$, $\alpha_3 = 0.2$).

4.1.2 Gender Equality

This measurement is aiming for making sure that both male and female have same opportunity for participating a specific sport and winning in the game. Two aspects are typically considered in this context: the ratio of male to female athletes participating in the event, and the distribution of medals between genders.

$$S_{\text{gen}} = \beta_1 \cdot (1 - |P_{\text{f}} - P_{\text{m}}|) + \beta_2 \cdot \left(1 - \frac{|W_{\text{f}} - W_{\text{m}}|}{W_{\text{t}}}\right) \tag{2}$$

where P_{f} is proportion of female athletes in the sport; P_{m} is proportion of male athletes in the sport; W_{f} is number of medals won by female athletes; W_{m} is number of medals won by male athletes; W_{t} is total number of medals in the event; and β_1, β_2 are weight coefficients, with $\beta_1 = 0.6$ and $\beta_2 = 0.4$.

4.1.3 Sustainability

Sustainability measures the inpact of Olimpic Games on environment, society, and economics. Sustainability principle is measured by carbon footprint, depletion of resources, and social impact of the campaign on the community.

$$S_{\text{sus}} = \gamma_1 \cdot \frac{E_{\text{e}}}{E_{\text{max}}} + \gamma_2 \cdot \frac{S_{\text{s1}}}{S_{\text{max1}}} + \gamma_3 \cdot \frac{C_{\text{e}}}{C_{\text{max}}} \tag{3}$$

where E_{e} is environmental protection score scaling from 0 and 1; S_{s1} is social responsibility score scaling from 0 and 1; C_{e} is economic efficiency score scaling from 0 and 1; E_{max} is maximum value of environmental protection score among all candidates; S_{max1} is maximum value of social responsibility score among all candidates; C_{max} is maximum value of economic efficiency score among all candidates; and γ_1, γ_2, γ_3 are weight coefficient, $\gamma_1 = 0.4$, $\gamma_2 = 0.3$, and $\gamma_3 = 0.3$.

4.1.4 Inclusivity

In order to keep the global multicultural appeal and comprehensive coverage of the Olympic events, the percentage of participation in each region and the number of participating countries were selected as indicators of inclusivity, which is calculated by following fomula.

$$S_{\text{inc}} = \delta_1 \cdot \frac{N_{\text{p}}}{N_{\text{Nmax,p}}} + \delta_2 \cdot \frac{C_{\text{d}}}{C_{\text{Nmax,d}}} \tag{4}$$

where N_{p} represents the total number of individuals worldwide who participate in the sport; C_{d} denotes the sports cultural diversity score, which ranges from 0 to 1. $N_{\text{Nmax,p}}$ indicates the highest level of global participation among all potential events, while $C_{\text{Nmax,d}}$ represents the highest cultural diversity score among these events. The weight coefficients δ_1 and δ_2 are set to 0.7 and 0.3, respectively.

4.1.5 Relevance and Innovation

In order to make sure the modernization of Olympic events and their appeal to younger spectators, the potential introduction of new technologies or rules was evaluated in terms of their impact on innovativeness. The percentage of young spectators was selected as a key indicator of the extent to which innovations would resonate with this demographic. An innovation evaluation index, which is calculated by the formula below, was constructed on a scale of 1 to 10 to assess the relative merits of different approaches.

$$S_{\text{rel}} = \phi_1 \cdot \frac{P_{\text{y}}}{P_{\text{max,y}}} + \phi_2 \cdot \frac{I_{\text{i}}}{I_{\text{max}}} \tag{5}$$

where P_{y} is proportion of young viewers aged 18-35; I_{i} is innovation score, ranging from 0 to 1, reflecting the introduction of new technologies and forms; $P_{\text{max,y}}$ is maximum proportion of young viewers among all candidate events; I_{max} is maximum innovation score among all candidate events δ_1 and δ_2 are weight coefficients, with $\phi_1 = 0.5$, $\phi_2 = 0.5$.

4.1.6 Safety and Fair Play

In order to guarantee the security of the athletes and the impartiality of the competition, the injury rate of the athletes and the fairness score (on a scale of 0-10) were selected as the parameters for measuring this indicator. The indicator is calculated as

$$S_{\text{saf}} = \theta_1 \cdot \frac{S_{\text{s2}}}{S_{\text{max2}}} + \theta_2 \cdot \frac{F_{\text{f}}}{F_{\text{max}}} \tag{6}$$

where S_{s2} is safety score ranging from 0 to 1; F_{f} is rating for anti-doping mechanism score ranging from 0 to 1; S_{max2} is maximum safety score among all candidate events; F_{max} is maximum anti-doping mechanism score among all candidate events; and θ_1, θ_2 are weight coefficients, with $\theta_1 = 0.6$ and $\theta_2 = 0.4$.

4.1.7 Economic Benefits and Commercialization Potential

In order to gauge the commercialization potential of the sport, its contribution to Olympic revenues, as well as the cultural value and global cultural impact of the program, a number of factors were selected as the measure for this indicator. These include commercial revenues, including tickets, sponsorships, and television broadcasts, advertising revenues, the number of sponsors, as well as the program's cultural legacy and global cultural impact. This indicator is measured by

$$S_{\text{eco}} = \kappa_1 \cdot \frac{R_{\text{r}}}{R_{\text{max}}} + \kappa_2 \cdot \frac{S_{\text{s3}}}{S_{\text{max3}}} \tag{7}$$

where R_{r} is the annual advertising revenue of the event (in millions of USD); S_{s3} is number of sponsors; R_{max} is maximum advertising revenue among all candidate events (in Millions of USD); S_{max3} is maximum number of sponsors among all candidate events; and κ_1, κ_2 are weight coefficients, with $\kappa_1 = 0.6$ and $\kappa_1 = 0.4$.

4.2 Calculated Evaluation Indicator for Several Sports

The table below shows the calculated evaluation indicators for several sparts in Olimpic Game.

Table 3: Result for Calculating Evaluation Indicators for Several Sports in Olympic Games

No.	Event Name	1	2	3	4	5	6	7
1	Water Sports	85.34	9.2	72.46	91.69	78.06	8.5	72.61
2	Archery	65.73	8.1	74.32	68.37	56.19	9.3	48.53
3	Athletics	99.21	9.8	87.51	95.09	89.42	9.7	85.43
4	Badminton	83.91	9.3	74.83	85.63	77.31	8.1	66.92
5	Baseball-Softball	81.08	8.6	67.23	80.39	70.14	7.5	72.77
6	Basketball	97.28	9.4	81.64	89.32	84.81	9.1	92.69
7	Basque Pelota	58.53	7.2	65.33	58.19	51.14	6.7	42.34
8	Boxing	68.72	7.4	63.46	64.35	59.79	7.9	50.06
9	Breakdancing	75.38	9.5	61.26	84.09	92.74	7.2	78.36
10	Canoeing	73.09	8.3	69.38	70.68	62.64	8.4	60.72
11	Cricket	62.34	7.9	71.83	66.38	69.18	7.1	50.28
12	Croquet	49.19	6.5	58.14	50.19	45.39	6	36.12
13	Cycling	92.71	9.1	90.16	80.47	84.69	8.9	85.01
14	Equestrian	81.37	9.2	62.53	75.34	68.43	9.3	55.66
15	Fencing	70.91	8	67.34	69.52	71.18	9.4	49.91
16	Field Hockey	74.47	8.1	78.92	71.33	62.48	8	61.64
17	Rugby	91.63	8.2	75.05	81.24	73.29	8.6	78.43
18	Football	99.08	9.7	85.01	96.39	88.24	9.6	100
19	Golf	74.29	8.4	65.21	70.12	71.89	8.2	61.53
20	Gymnastics	82.18	9.9	79.14	76.53	79.74	9	72.39
21	Handball	78.45	8.3	72.18	75.68	70.23	8.1	63.29
22	Tennis	85.92	9.1	74.63	80.21	83.14	8.7	82.19
23	Judo	72.37	8.8	66.54	69.87	64.38	8.4	58.26
24	Karate	69.14	8.6	63.49	67.59	61.72	7.9	54.61
25	Lacrosse	66.42	7.7	64.18	65.23	60.49	7.6	53.14
26	Modern Pentathlon	70.58	8	68.34	66.85	67.29	7.9	56.73
27	Polo	60.18	7.1	62.72	60.94	57.46	6.9	49.38
28	Squash	73.86	8.2	67.59	69.52	75.43	8	65.12
29	Roque	50.29	6.4	57.36	51.68	46.72	6.1	37.54

Continued on next page

No.	Event Name	1	2	3	4	5	6	7
30	Rowing	76.12	8.5	73.61	72.47	70.93	8.3	63.91
31	Rugby	84.67	8.3	70.59	78.29	72.42	8.5	67.83
32	Sailing	70.42	8.2	72.49	68.73	67.18	7.8	59.73
33	Shooting	68.29	7.9	66.38	64.23	63.74	8	55.81
34	Skateboarding	77.14	9.3	68.27	74.38	83.51	7.4	73.14
35	Sport Climbing	75.39	9.2	69.14	71.62	80.82	8.1	68.91
36	Squash	71.86	8.5	65.24	69.03	76.18	7.9	64.27
37	Surfing	74.19	8.7	66.58	73.42	82.41	7.6	70.29
38	Table Tennis	83.42	9.1	73.12	79.48	77.23	8.3	66.47
39	Taekwondo	72.59	8.9	68.14	69.73	69.52	8	60.29
40	Tennis	85.92	9.2	74.63	80.21	83.14	8.8	82.19
41	Triathlon	80.13	8.6	70.85	73.76	74.67	8.4	67.58
42	Tug of War	59.76	6.8	60.12	62.37	52.81	6.5	49.27
43	Volleyball	90.28	9.4	80.32	85.27	86.49	9.1	91.04
44	Water Motor Sports	63.49	7.5	64.13	67.53	61.21	7.3	55.78
45	Weightlifting	68.21	7.9	66.28	64.83	63.41	7.8	55.24
46	Wrestling	74.68	8.3	70.43	72.21	70.89	8.2	64.28
47	Skating	79.37	8.8	71.47	75.53	73.52	8.4	66.12
48	Ice Hockey	82.14	8.9	73.52	77.69	75.82	9	69.93

5 BUIDING MODEL BY USING COMBINE WEIGHTING METHOD

To help the IOC provide a reliable, multi-dimensional evaluation model to help select the best sustainable development initiatives that meet Olympic standards, the modeling will use the Combined Weighting Method (CWM) to determine the weight of each evaluation indicator [4]. Specifically, we will obtain the initial weights through the analytic hierarchy process (AHP), entropy weight method and coefficient of variation method, and construct the weight matrix. Then, calculate the Kendall correlation coefficient and perform a consistency test, and then use the CRITIC method to weight the weight matrix to ensure that each weight can reflect both subjective judgment and objective differences. At the same time, the MC-TOPSIS (Monte Carlo Simulation-TOPSIS) model is used for comparison. The final weight of each SDE is applied to Monte Carlo simulation, and multiple scenarios are generated through a large number of random samples to examine the ranking stability of SDE under different uncertainty conditions. This process enables identification of the best performing SDEs under various possible scenarios and assessment of their performance differences across weight fluctuations. The flow chart for this model is drawn in Figure 3.

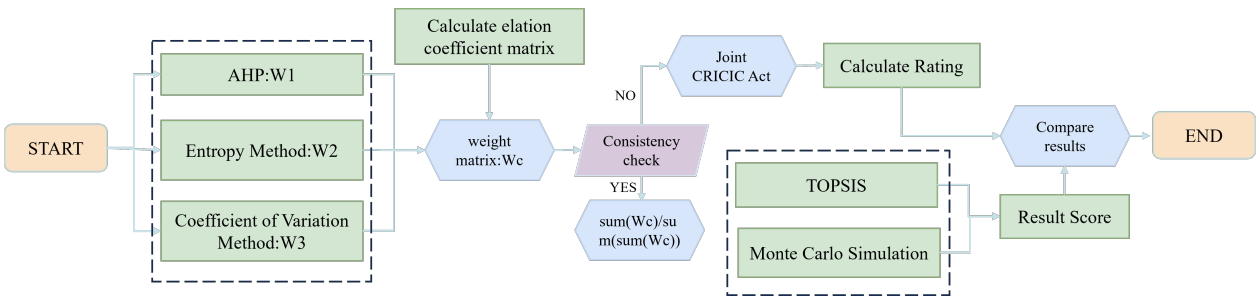


Figure 3: Weight Anaysis Model Construction

5.1 Standardized Scoring

Since evaluation indicators of different criteria varies in units and range, it is essential to standardize all the data to ensure they are on the same scale before follow data processing, making easier for subsequent analysis and comparison. Z-score Standardization [5], in this case, can be utilized–the Z score for each score is calculated, and the standardized value is:

$$z_i = \frac{x_i - \mu}{\sigma} \tag{8}$$

where μ is the mean and σ is the standard deviation.

5.2 Deriving Weight Matrix

After standarlizing the score, the model use Combined Weighting Method to give weight to each evaluation indicator. It combinded three method: analytic hierarchy process, entropy weight method, and coefficient of variation method.

5.2.1 Analytic Hierarchy Process (AHP)

Analytic Hierarchy Process is a decision-making method used to judge the relative importance of factors. In this model, the relative importance of each factor is first judged by expert review, a comparison matrix is constructed and weights are calculated.

STEP 1: Constructing the Judgment Matrix A**.** Create a $n \times n$ judgment matrix $A = \left[a_{ij}\right]$, where a_{ij} denotes the ratio of importance between factor i and factor j. It is common to use a scale of 1-9 to indicate the level of importance:

Table 4: Pairwise Comparison Scale for Factors

Value	Meaning
$a_{ij} = 1$	The two factors are equally important.
$a_{ij} = 3$	Factor i is slightly more important than factor j.
$a_{ij} = 5$	Factor i is clearly more important than factor j.
$a_{ij} = 7$	Factor i is much more important than factor j.
$a_{ij} = 9$	Factor i is extremely more important than factor j.
$a_{ij} = \frac{1}{a_{ji}}$	The importance of factor j relative to factor i.

STEP 2: Calculate the Eigenvector and Eigenvalue. Compute the eigenvector $T = [t_1, t_2, \cdots, t_n]^T$ and the maximum eigenvalue $\lambda_{\max}$ of the judgment matrix A. The eigenvector T represents the relative weight of each factor and the maximum eigenvalue $\lambda_{\max}$ is used for consistency checking.

STEP 3: Calculate the Normalized Weight Vector. Normalize each column of the judgment matrix A to obtain the weight vector $W_1 = [w_{11}, w_{12}, \cdots, w_{1n}]^T$ where each weight w_{1i} is calculated by the following formula:

$$w_{1i} = \frac{t_i}{\Sigma_{j=1}^{n} t_j} \tag{9}$$

where t_i is the i-th component of the eigenvector T.

STEP 4: Consistency Check. Consistency index CI and the consistency ratio CR is calculated by following fomula:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad \text{and} \quad CR = \frac{CI}{RI} \tag{10}$$

where RI is a stochastic consistency index that depends on the order n of the matrix and can be found in the following table:

Table 5: Random Index (RI) Values for Different Matrix Sizes (n)

n	1	2	3	4	5	6	7	8	9
RI	0.0	0.0	0.58	0.90	1.12	1.24	1.32	1.41	1.45

If $CR < 0.1$, the judgment matrix passes the consistency test, otherwise the judgment matrix needs to be adjusted. Ultimately, if the consistency test passes, the weight vector W_1 is the final weight of each factor.

5.2.2 Entropy Method

The entropy weighting method determines the weights by calculating the entropy value of the factors, which reflects the uncertainty of the data, and the larger the entropy value, the smaller the weights.

STEP 1: Calculate the Proportion of Each Indicator. After normalization, we calculate the proportion of each evaluation object on each indicator p_{ij}:

$$p_{ij} = \frac{y_{ij}}{\Sigma_{i=1}^{m} y_{ij}} \tag{11}$$

where p_{ij} is the proportion of the i-th evaluation object on the j-th indicator; and $\Sigma_{i=1}^{m} p_{ij} = 1$ guarantees that the the proportions for each indicator sum to 1.

STEP 2: Calculate the Entropy E_j. According to the formula of information entropy, calculate the entropy E_j value of each indicator:

$$E_j = -k \sum_{i=1}^{m} p_{ij} \ln\left(p_{ij}\right) \tag{12}$$

where $k = 1/\ln(m)$ is a constant used to ensure that the entropy E_j is between $[0, 1]$.

STEP 3: Calculate the Degree of Divergence d_j. The degree of divergence d_j represents the information provided by each indicator, and is defined as:

$$d_j = 1 - E_j \tag{13}$$

STEP 4: Calculate the Weight. Based on the degree of divergence d_j, we calculate the weight vector $W_2 = [w_{21}, w_{22}, \cdots, w_{2n}]^T$ for each indicator by

$$w_{2j} = \frac{d_j}{\Sigma_{j=1}^{n} d_j} \tag{14}$$

5.2.3 Coefficient of Variation Method

The coefficient of variation method determines the weights based on the dispersion of each indicator; the higher the coefficient of variation, the more important the indicator is in the overall decision-making process.

STEP 1: Calculate the Mean μ_j and Standard Deviation σ_j for Each Indicator. For each indicator j, calculate the mean μ_j and standard deviation σ_j for the standardized data y_{ij}:

$$\mu_j = \frac{1}{m}\sum_{i=1}^{m} y_{ij} \quad \text{and} \quad \sigma_j = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(y_{ij} - \mu_j\right)^2} \tag{15}$$

STEP 2: Calculate the Coefficient of Variation CV_j. The coefficient of variation CV_j is used to measure the dispersion of each indicator:

$$CV_j = \frac{\sigma_j}{\mu_j} \tag{16}$$

STEP 3: Calculate the Weight W_3. The coefficient of variation CV_j was used to compute the weights of each indicator vector vector $W_3 = [w_{31}, w_{32}, \cdots, w_{3n}]^T$:

$$w_{3j} = \frac{CV_j}{\Sigma_{j=1}^{n} CV_j} \tag{17}$$

By integrating the weights from different methods, weighted matrix $W_c = \left[w_{c,ij}\right]_{3\times n}$ can be obtained; that is

$$W_c = \begin{bmatrix} W_1 & W_2 & W_3 \end{bmatrix} \tag{18}$$

5.3 Determination of Final Weight Coefficients

5.3.1 Calculation of the Kendall Correlation Coefficient Matrix

The Kendall correlation coefficient (Kendall's tau coefficient) is used to measure the degree of ranking consistency between different weight vectors [6]. Here, the Kendall's correlation coefficients between each weight vector W_1, W_2, and W_3 are calculated to understand the consistency between them. The specific calculation is as follows:

1. Let W_i and W_j be the weight vectors obtained from two different evaluation methods.
2. Calculate the Kendall correlation coefficient τ_{ij} by using fomula

$$\tau_{ij} = \frac{2\left(S^+ - S^-\right)}{n\left(n-1\right)} \tag{19}$$

   where S^+ is the number of consistently ordered pairs in the weight vector; S^- is the number of inconsistently ordered pairs in the weight vector; and n is the total number of factors.
3. The correlation coefficients of each pair of weight vectors were deposited into the Kendall correlation coefficients matrix R_2 and the NaN value in R_2 was replaced with 0 to avoid the interference of missing data.

5.3.2 Consistency Test

In order to verify the consistency between the weight vectors, a consistency test is performed using the sum of the Kendall correlation coefficient matrix. The specific steps are as follows:

1. Extract the first row of the Kendall correlation coefficient matrix R_2, ignoring the diagonal elements, to form the vector $R = R_2\left(1, 2 : \text{end}\right)$.
2. Calculate the sum of R and compare it with $0.05 \times \text{length}\left(R\right)$ to determine whether the consistency condition is met:

$$\sum\left(R\right) < 0.05 \times \text{length}\left(R\right)$$

3. If the above conditions hold, the consistency test is considered passed. The final weight coefficients vector $W_{\text{final}} = \left[w_{\text{final},1}, w_{\text{final},2}, \cdots, w_{\text{final},n}\right]$ of the computed weight matrix W_c can be derived where

$$W_{\text{final},j} = \frac{\sum_{i=1}^{3} W_{c,ij}}{\sum_{i=1}^{3}\sum_{j=1}^{n} W_{c,ij}} \tag{20}$$

or, if the above conditions doesn't hold, the CRICIC method will be used to recalculate.

5.3.3 The CRICIC Method

STEP 1: Calculation of Contradiction Standard Deviation and Information Capacity.

1. **Calculation of Weighted Contradiction.** Calculate the standard deviation σ of each weight vector W_i $(i = 1, 2, 3)$ by using fomula

$$\sigma(W_i) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(W_i - \overline{W}_i\right)^2} \tag{21}$$

Thus, the standard deviation vector for W_c can be defined as $\sigma(W_c) = [\sigma(W_1), \sigma(W_2), \sigma(W_3)]$.

2. **Calculation of Information Capacity.** Calculate the complementarity of the Kendall correlation matrix R, that is, the information capacity f:

$$f = \sum_{i=1}^{n}(1 - R_i) \tag{22}$$

STEP 2: Calculation of Weighted Contradiction and Weight Normalization.

1. **Calculation of Weighted Contradiction.** Calculate the weighted contradiction coefficient vector $C = [c_1, c_2, c_3]$, that is, the product of the standard deviation matrix $\sigma(W_c)$ and the information capacity f:

$$C = f \cdot \sigma(W_c) \tag{23}$$

2. **Weight Normalization.** To normalize weighted contradiction coefficient vector C to obtain the new weight coefficients vector $W_{\text{new}} = \left[w_{\text{new},1}, w_{\text{new},2}, w_{\text{new},3}\right]$ where

$$w_{\text{new},i} = \frac{c_i}{\sum_{i=1}^{3} c_i} \tag{24}$$

STEP 3: Recalculation of Comprehensive Weights and Result Validation.

1. **Recalculation of Comprehensive Weights.** Use the new weight coefficients vector W_{new} to reweight and obtain the final weight vector W_{final}:

$$W_{\text{final}} = W_{\text{new}} \cdot W_c \tag{25}$$

2. **Result Validation.** Apply W_{final} to the comprehensive evaluation calculation to ensure the stability and rationality of the results.

5.4 Model Output

After obtaining the final weight coefficient W_{final}, they can be applied to the data matrix A for weighted summation to calculate the score S:

$$S = A \cdot W_{\text{final}}^{T} \tag{26}$$

The weight coefficient obtained through the above method can be calculated as the comprehensive score. Performance is ranked accoring to the calculated score to ultimately determine which sports best meet the IOC's standards. The results are shown in the following table. The serial numbers in this table correspond to the ranking serial numbers of the sports in the attachment given in the question.

Table 6: Scores for Different Sports Events in the Olympic Games

Ranking	Serial No.	Event	Score	Ranking	Serial No.	Event	Score
1	3	Athletics	100.000	25	19	Golf	50.462
2	18	Football	99.464	26	47	Skating	49.255
3	6	Basketball	91.301	27	23	Judo	48.495
4	13	Cycling	87.620	28	15	Fencing	48.156
5	43	Volleyball	85.441	29	10	Canoeing	47.830
6	40	Tennis	75.452	30	32	Sailing	46.802
7	20	Gymnastics	75.089	31	26	Modern Pentathlon	44.005
8	1	Aquatics	72.392	32	46	Wrestling	43.605
9	4	Badminton	70.113	33	2	Archery	41.490
10	38	Table Tennis	67.588	34	24	Karate	41.239
11	9	Breaking	63.885	35	45	Weightlifting	40.103
12	31	Rugby	62.652	36	33	Shooting	39.453
13	34	Skateboarding	62.160	37	11	Cricket	37.300
14	48	Ice Hockey	61.065	38	8	Boxing	34.193
15	41	Triathlon	60.591	39	25	Lacrosse	34.146
16	35	Sport Climbing	60.406	40	44	Water Motorsports	32.177
17	14	Equestrian	59.114	41	17	Flag Football	31.786
18	5	Baseball and Softball	57.747	42	27	Polo	22.781
19	37	Surfing	56.335	43	7	Tug of War	17.751
20	21	Handball	56.214	44	42	Tug de Paume	4.212
21	30	Rowing	56.181	45	22	Jue de Paume	6.748
22	36	Squash	51.488	46	28	Rackets	4.212
23	16	Feild Hockey	51.282	47	29	Roque	1.978
24	39	Tackwondo	51.032	48	12	Croquet	0.828

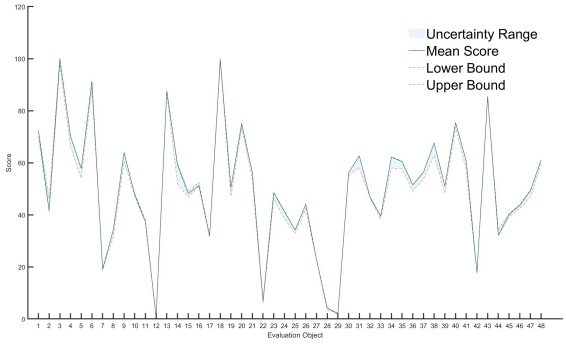


Figure 4: Evaluation Score Distribution 1

5.5 Model Comparison

The MC-TOPSIS (Monte Carlo Simulation-TOPSIS) model is introduced for comparison [7]. MC-TOPSIS is a multi-criteria decision-making method that combines Monte Carlo simulation with the TOPSIS method. TOPSIS is a system based on ideal points and negative.The sorting method of ideal points is sorted by calculating the distance between each solution and the ideal solution and the negative ideal solution. The core of MC-TOPSIS is to generate multiple possible scenarios through Monte Carlo simulation, perform multiple random perturbations on the input data to better consider the randomness or uncertainty of the input data, and finally perform statistical analysis on the ranking of decision solutions.

5.5.1 Defining Ideal and Negative Ideal Solutions

In TOPSIS, the positive ideal solution (A^+) and the negative ideal solution (A^-) are defined as follows:

1. The positive ideal solution A^+ is the vector of the optimal values for all criteria, denoted as $A^+ = \left\{y_1^+, y_2^+, \cdots, y_n^+\right\}$;

2. The negative ideal solution A^- is the vector of the optimal values for all criteria, denoted as $A^- = \left\{y_1^-, y_2^-, \cdots, y_n^-\right\}$.

5.5.2 Monte Carlo Simulation

Monte Carlo simulations were performed on the data for each assessment object to generate a range of possible scenarios to deal with the uncertainty in the data:

STEP 1: Determine the random disturbance range. Based on the data characteristics (e.g., mean and variance) of each evaluation indicator, determine the data perturbation range, which can usually be assumed to follow a normal or uniform distribution.

STEP 2: Generate simulation samples. Apply multiple random disturbances to each index value y_{ij}, generating N sets of simulation samples $\left\{Y_{ij}^1, Y_{ij}^2, \cdots, Y_{ij}^n\right\}$, where Y_{ij}^k represents the value of the i-th evaluation object on the j-th index in the k-th simulation.

STEP 3: Rebuild the evaluation matrix. The evaluation matrix generated for each simulation can be expressed as $Y^k = \left[Y_{ij}^k\right]$, where $k = 1, 2, \cdots, N$.

5.5.3 Calculate the TOPSIS Score for Each Simulation

Perform the TOPSIS calculations for each evaluation matrx Y^k generated by the sitimulation as follow procedure.

STEP 1: Standardize the Evaluation Matrix. Standardize Y^k to obatain the standardzed matrix $R^k = \left[r_{ij}^k\right]$, where

$$r_{ij}^k = \frac{Y_{ij}^k}{\sqrt{\Sigma_{i=1}^m \left(Y_{ij}^k\right)^2}} \tag{27}$$

STEP 2: Weoighted Standardize Matrix. To obtain the weighted matrix $V^k = \left[v_{ij}^k\right]$, multiplying the weight W to the standardized matrix R^k, where

$$v_{ij}^k = w_j \cdot r_{ij}^k \tag{28}$$

STEP 3: Calculate the Distances to the Positive and Negative ideal solutions.

1. The positive and negative ideal solation, A^+ and A^- respectively, are defined as

$$\begin{aligned} A^+ &= \left\{\max\left(v_{1j}^k\right), \max\left(v_{2j}^k\right), \cdots, \max\left(v_{nj}^k\right)\right\} \\ A^- &= \left\{\min\left(v_{1j}^k\right), \min\left(v_{2j}^k\right), \cdots, \min\left(v_{nj}^k\right)\right\} \end{aligned} \tag{29}$$

2. Calculate the Euclidean distance of each evaluation object to the ideal and negative ideal solutions; those are

$$\begin{aligned} D_i^{+k} &= \sqrt{\sum_{j=1}^n \left(v_{ij}^k - y_j^+\right)^2} \\ D_i^{-k} &= \sqrt{\sum_{j=1}^n \left(v_{ij}^k - y_j^-\right)^2} \end{aligned} \tag{30}$$

STEP 4: Calculate the relative closeness C_i^k. The relative closeness C_i^k of the i-th evaluation object in the k-th simulation is given by

$$C_i^k = \frac{D_i^{-k}}{D_i^{+k} + D_i^{-k}} \tag{31}$$

5.5.4 Statistical Analysis of the Results from Multiple Simulations

Repeat the above steps for N simulations to obtain the relative proximity C_i^k, of each evaluation object in each simulation, and then statistically analyze the relative proximity of each evaluation object to calculate its mean and standard deviation to obtain more stable results. The average closeness for each evaluation object can be calculated by

$$\bar{C}_i = \frac{1}{N} \sum_{k=1}^{N} C_i^k \tag{32}$$

and the standard deviantion of the closeness for each evaluation object σ_{C_i} is calculated by

$$\sigma_{\bar{C}_i} = \sqrt{\frac{1}{N-1} \sum_{k=1}^{N} \left(C_i^k - \bar{C}_i\right)^2} \tag{33}$$

5.5.5 Ranking and Result Analysis

All evaluation objects are ranked according to the average proximity $\bar{C}_i$. The higher the proximity, the higher the ranking. The standard deviation $\sigma_{\bar{C}_i}$ of the proximity can be used as a measure of the stability of the ranking results; the smaller the standard deviation, the more stable the ranking. The results can be seen. Upon comparison, it is found that the results are similar. However, the MC-TOPSIS method has a larger error band, indicating that the CWM model yields better results:

Table 7: MC-Topsis Ranking for Sports

Ranking	Event	Score	Ranking	Event	Score
1	Football	98.936430	25	Golf	47.953314
2	Athletics	96.861979	26	Canoeing	47.861390
3	Basketball	89.616694	27	Fencing	47.731547
4	Cycling	85.993677	28	Skating	47.550199
5	Volleyball	85.768341	29	Sailing	47.307150
6	Tennis	73.695559	30	Judo	47.976280
7	Gymnastics	72.723272	31	Archery	46.172152
8	Aquatics	69.875080	32	Wrestling	43.333822
9	Badminton	66.882912	33	Modern Pentathlon	42.866223
10	Table Tennis	63.737268	34	Weightlifting	40.095087
11	Ice Hockey	59.895655	35	Karate	39.672697
12	Rugby	59.366788	36	Cricket	38.979617
13	Breaking	59.318223	37	Shooting	38.923704
14	Skateboarding	58.460085	38	Water Motorsports	34.248487
15	Triathlon	58.063067	39	Lacrosse	33.593957
16	Sport Climbing	57.730372	40	Boxing	32.883900
17	Rowing	55.683576	41	Flag Football	32.769615
18	Baseball and Softball	55.434577	42	Polo	23.117768
19	Handbal	54.791165	43	Basque Pelota	19.683116
20	surfing	54.320044	44	Tug of War	18.811518
21	Field Hockey	53.189406	45	Jeu de Paume	6.883763
22	Equestrian	52.737058	46	Rackets	4.919216
23	Squash	49.971049	47	Roque	2.369794
24	Taekwondo	49.007374	48	Croquet	1.726015

6 Analysis of Changes in Olympic SEDs

6.1 Selection of Test Sports

In this part, three sports events (SDEs) are seleted to be either added or removed in the 2020, 2024, and 2028 Olimpic Games; they are

1. **Karate.** Karate is assumed to be added in the 2020 Tokyo Olympics.
2. **Sport Climbing.** Sport climbing is assumed to be added in the 2024 Tokyo Olympics.
3. **Offshore Rowing.** Offshore rowing is assumed to be added in the 2028 Tokyo Olympics.

Table 8: Sport Events Rating Table 1

Attribute	Karate	Sport Climbing	Offshore Rowing
Popularity and Accessibility	69.14	73.39	63.27
Gender Equality	8.6	9.2	7.5
Sustainability	63.49	69.14	72.36
Inclusivity	67.59	71.62	65.47
Relevance and Innovation	61.72	80.82	60.94
Safety and Fair Competition	7.9	8.1	7.1
Economic Benefits and Commercialization Potential	54.61	68.91	51.38

Next, selected 6 events from the previous 20 categories of sports as input data for the next step of evaluation, the calculated results are shown below:

Table 9: Sport Event Rating Table 2

Attribute	Athletics	Archery	Basketball	Gymnastics	Boxing	Aquatics
Popularity and Accessibility	99.21	65.73	97.28	82.18	68.72	85.34
Gender Equality	9.8	8.1	9.4	9.9	7.4	9.2
Sustainability	87.51	74.32	81.64	79.14	63.46	72.46
Inclusivity	95.09	68.37	89.32	76.53	64.35	91.68
Relevance and Innovation	89.42	56.19	84.81	79.74	59.79	78.06
Safety and Fair Competition	9.7	9.3	9.1	9.0	7.9	8.5
Economic Benefits and Commercialization Potential	85.43	48.53	92.63	72.39	50.06	72.61

6.2 Calculation of Scores

Applying the combined assignment comprehensive evaluation model to continue the evaluation and analysis, the initial weights W_1, W_2, W_3 were obtained through the hierarchical analysis method (AHP), entropy weight method and coefficient of variation method methods, and the weight matrix W_c was constructed, and at the

same time, CRITIC method to get the weights W_{final}:

$$W_c = \begin{bmatrix} 0.3899 & 0.1470 & 0.1508 \\ 0.1666 & 0.1087 & 0.1161 \\ 0.0949 & 0.1333 & 0.1343 \\ 0.0563 & 0.1807 & 0.1690 \\ 0.1824 & 0.1287 & 0.1333 \\ 0.0734 & 0.1360 & 0.1408 \\ 0.0366 & 0.1658 & 0.1559 \end{bmatrix} \quad \text{and} \quad W_{\text{final}} = \begin{bmatrix} 0.8522 & 0.0894 & 0.0583 \end{bmatrix}$$

The results are calculated as follow:

Table 10: Score Table of Sports Events

No.	**4**	**6**	**7**	**9**	**2**	**5**	**1**	**8**	**3**
Event	Athletics	Basketball	Gymnastics	Aquatic Sports	Sport Climbing	Archery	Karate	Boxing	Offshore Rowing
Score	100	87.678	64.611	61.825	44.004	19.375	18.11	8.5537	7.6933

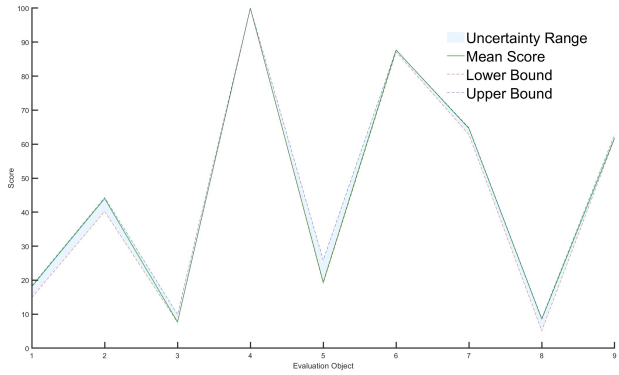


Figure 5: Evaluation Score Distribution 2

6.3 Results Analysis

Karate, **Sport Climbing**, and **Offshore Rowing** have exhibited diverse performances, as newly added or soon-to-be-added events in recent years. Karate has shown good popularity and inclusivity, but due to a lack of innovation, its ability to attract young audiences is limited, resulting in weaker market appeal. Sport Climbing stands out in terms of innovation, gender equality, and sustainability, aligning with the diversified needs of modern sports. Offshore Rowing excels in sustainability, yet its popularity and economic benefits are relatively weak, with room for improvement in global audience recognition and commercial value. Overall, these events have advantages in some indicators but face challenges in terms of global audience reach and marketization.

Athletics, **Archery**, **Basketball**, **Gymnastics**, **Boxing**, and **Aquatic Sports (Swimming)** are traditional Olympic disciplines with a stable audience base and global participation. Track and field and basketball excel in popularity, inclusivity, gender equality, and economic benefits, thereby consolidating their core status. Gymnastics and aquatic sports enjoy good popularity, yet there is still room for improvement in sustainability. Archery and boxing perform well in terms of gender equality and safety, but their popularity

and economic benefits are relatively low, leading to insufficient market promotion and commercialization potential. Overall, traditional disciplines possess stability and a broad audience base, but they need to strengthen their marketization to maintain their competitiveness.

The newly added or removed events (**Karate**, **Sport Climbing**, and **Offshore Rowing**) have advantages in innovation and sustainability, aligning with the Olympic Games' environmental protection trend and diversified development. However, they lack a strong global audience base and commercialization. Traditional events (**Athletics**, **Archery**, **Basketball**, **Gymnastics**, **Boxing**, and **Aquatic Sports (Simming)**) have consolidated their status through stability and high global participation, but some of them need to enhance their market appeal. Overall, while maintaining the Olympic spirit and the value of traditional events, the Olympic Games must strike a balance between tradition and innovation to attract new audiences.

7 Selection of New and Potential Projects

7.1 Additions or Reintroductions SDEs in 2032

When evaluating the potential new or reintroduced sports disciplines (SDEs) for the 2032 Brisbane Olympics, we comprehensively considered factors such as popularity, inclusivity, sustainability, economic benefits, and audience appeal. Below are our top three recommended projects, along with their order of priority:

Table 11: Project Data Table

Project	Audience Base (millions)	Countries Participating	Economic Market Value (USD billions)
Cricket	2500.5	105	62.1
Martial Arts	500	128	8.3
Squash	20.4	40	1.2

First Recommendation: Cricket. With approximately 2.5 billion cricket enthusiasts worldwide, particularly in countries like India, Pakistan, the United Kingdom, and Australia, this vast audience base holds immense potential for drawing significant attention to the Olympics. Cricket events attract substantial commercial sponsorship, and T20 cricket, due to its fast pace and high entertainment value, is well-suited for the Olympic schedule. T20 cricket is relatively environmentally friendly and has moderate requirements for venues and event scheduling, making it suitable for Olympic conditions. It also appeals to younger audiences.

Second Recommendation: Martial Arts. As a traditional Asian sport, martial arts has a high level of global promotion, with diverse event formats such as Wushu routines and Sanshou (free combat). It particularly appeals to youth groups and is gender-equal, aligning with the values of the Olympics. According to data from the International Wushu Federation, martial arts activities have been officially conducted in over 150 countries and regions worldwide, with more than 80 million registered enthusiasts, demonstrating significant global attention. Martial arts events have low venue requirements and can be flexibly arranged using existing facilities, thus saving costs. The promotion of martial arts events is mature, with widespread global participation, particularly in Europe and North America, where large-scale international events are held annually.

Third Recommendation: Squash. Squash is popular globally and, as a high-intensity competitive sport, is particularly favored by younger audiences. According to statistics, there are over 20 million squash enthusiasts worldwide, distributed across 185 countries and regions. The viewing and competitive nature of squash events has been continuously enhanced, with squash at the Commonwealth Games attracting millions of television viewers, demonstrating its appeal to younger audiences. Squash courts occupy a small area and have high resource utilization, aligning with the sustainable development needs of the Olympics.

Squash events are widely conducted globally, including the PSA World Tour, showcasing high inclusivity and participation. Squash courts have low requirements, with low setup costs and easy recyclability, aligning with the Olympic spirit. It is suitable as an emerging Olympic sport.

7.2 Potential SDEs for 2036

For the Olympics in 2036 and beyond, we recommend the following potential projects. These projects have strong appeal among younger audiences and align with the development trends of the Olympics.

Esports. Esports has a vast young audience base globally. According to Newzoo, the global esports audience is expected to reach 520 million people in 2024, and the total revenue of the global esports market in 2022 reached 1.2 billion, primarily from event sponsorship, advertising, and live streaming. Esports possesses competitiveness in tactics, teamwork, and hand speed(Actions Per Minute, APM) reaction, and is gradually being recognized by the sports community [8]. However, its inclusion in the Olympics would require consideration of sustainability, fairness of viewing, and its impact on the health of teenagers. Esports venues are flexible, and the equipment is convenient, suitable for settings of various sizes. The global esports infrastructure is well-developed, and events can be broadcast on multiple platforms, possessing wide audience appeal and economic value. It is a potential project for future Olympics.

Ultimate Frisbee. Ultimate frisbee is popular in North America and Europe, with simple rules and high viewing appeal, making it favored by young people and students. According to reports from the World Flying Disc Federation, there are approximately 7.5 million enthusiasts worldwide, distributed across 85 countries. Ultimate Frisbee emphasizes teamwork, sportsmanship, and "self-refereeing", aligning with the Olympic ideal of fair competition. Ultimate Frisbee has low venue requirements, is environmentally friendly, and has low facility costs. It has high potential for promotion in intercollegiate and community events, aligning with the Olympics' goal of promoting sustainable sports projects.

Beach Rugby. Beach rugby is a simplified version of rugby with a fast pace, strong confrontation, and extremely high viewing appeal. Data shows that there are over 270 million rugby enthusiasts worldwide, and beach rugby is gradually becoming popular among younger audiences, suitable for short viewing durations. Beach rugby is played on outdoor venues such as beaches, with low facility requirements and maintenance costs, making it suitable for Olympic Games held in coastal cities. The events are flexible and highly entertaining, enriching the beach-themed events at the Olympics.

8 SENSITIVITY ANALYSIS

8.1 Model Robustness Evaluation

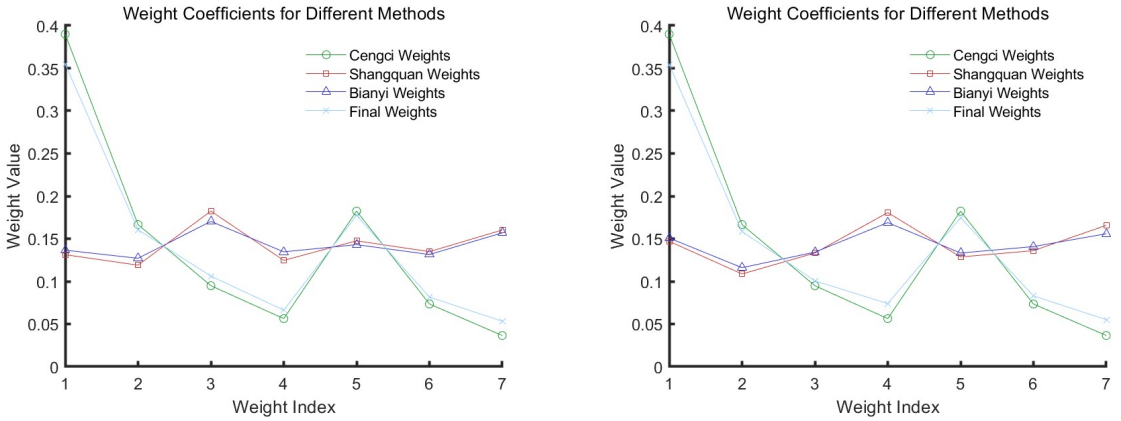
In this model, we have adopted a multi-criteria decision-making framework, incorporating comprehensive weighting methods such as the Analytic Hierarchy Process (AHP), Entropy Weight Method, Coefficient of Variation Method, and CRITIC method, along with the MC-TOPSIS model. These methods collectively construct a multi-dimensional and highly comprehensive scoring system, effectively balancing objectivity and flexibility.

8.1.1 Robustness Performance

Multi-Indicator Coverage. The model covers multiple dimensions such as popularity, gender equality, sustainability, inclusivity, innovation, fairness, and economic benefits, capable of reflecting the comprehensive value of SDEs from various aspects. The weights of multiple indicators are derived through multiple weighting methods, reducing the risk of bias.

Uncertainty Handling. MC-TOPSIS introduces Monte Carlo simulation to simulate multiple scenarios,taking into account fluctuations in different weights and changes in scores. This can reduce the extreme impact of a single score and make the final result more stable, demonstrates higher sensitivity.

Comparison of Indicator Weight. We found that the weight values of indicators 1, 4, and 7 vary significantly across different methods, indicating that these three types of indicators have a relatively large impact on the results.



(a) Weight Coefficients for Different Methods 1 (b) Weight Coefficients for Different Methods 2

Figure 6: Weight Coefficients for Different Methods

8.1.2 Potential Uncertainties

Subjectivity in Weight Allocation. Even with weight allocation through methods such as AHP and CRITIC, there may still be influence from experts' subjective opinions.

Data Dependence. The output results of the model are highly dependent on the accuracy and update rate of the data, especially those related to popularity and commercialization potential, which often fluctuate due to regional differences and variations in audience demographics.

Robustness Conclusion. In summary, the model demonstrates high robustness and fault tolerance, effectively adapting to variations in different SDEs data. However, in terms of specific weights and data dependence, the model may be influenced by certain subjective and regional factors. Continuous adjustment and updating of data during use are required to ensure the scientific validity of decision-making.

8.2 Analysis of the Advantages and Disadvantages of Factors with Higher Scores

In the SDEs model, factors contributing to higher scores include popularity and accessibility, inclusivity, economic benefits and commercialization potential, as well as relevance and innovation. Below is an analysis of the advantages and disadvantages of these factors.

Table 12: Advantages and Disadvantages of Factors with Higher Scores

Factors	Advantages	Disadvantages
Popularity and Accessibility	Projects with high popularity attract a large audience, which is beneficial for the global promotion of the Olympics. This indicator is easily quantifiable through data such as global audience size and event coverage, providing an intuitive reflection of the project's global influence.	Popularity varies significantly across different regions, and some projects may lack global appeal, potentially overlooking niche but culturally valuable projects.
Inclusivity	Projects with high inclusivity reflect multiculturalism and international participation, aligning with the Olympic spirit and enhancing the global representativeness of the Olympics.	Projects with high inclusivity may not necessarily have widespread global commercial value, which could affect the Olympics' revenue and sponsorship appeal.
Economic Benefit and commercial Potential	Projects with strong commercialization potential attract sponsorship, contributing to enhanced economic benefits, and increase visibility through media coverage.	Over-commercialization may diminish the cultural value of the Olympics, causing niche projects to be overlooked, and affecting the diversity and balance of the Olympics.
Relevance and Innovation	Innovative projects attract younger audiences, adding vitality and staying in line with modern trends, aligning with the trend of introducing emerging projects.	Innovative projects may face safety and adaptability issues, with some project rules still not fully developed, and global recognition may take time to build up.

8.3 Conclusion

Overall, the factors in the model that contribute to higher SDEs scores play a positive role in the diversification, inclusivity, and global influence of the Olympics. However, commercial factors such as popularity and economic benefits sometimes conflict with the goals of balance and diversity in the Olympic spirit [9]. To optimize future decision-making, it is recommended to introduce innovative projects moderately while retaining mainstream projects, in order to maintain the global appeal and diversity of the Olympics. Additionally, as the hosting locations of the Olympics and global sports trends change, dynamically adjust the weights in the model to enhance its robustness and adaptability.

LETTER

From: HiMCM Olympic Consultants (HOC)
To: International Olympic Committee
Subject: Research on Olympic Event Selection Model Based on Combined Weighting Method (CWM)

Dear International Olympic Committee,

In our research on the selection of Olympic sports events, we have adopted a model based on the Combined Weighting Method (CWM). By fusing multiple weighting methods and analyzing the dynamic changes of Olympic sports events (SDEs), we have found that traditional Olympic sports such as athletics, archery, basketball, gymnastics, boxing, and aquatic sports continue to occupy a core position due to their solid audience base and widespread global participation. At the same time, emerging sports such as karate, sport climbing, and offshore rowing, while performing well on certain specific indicators, still face significant challenges in terms of global audience base and marketability.

Based on the comprehensive evaluation of seven factors, we have assessed the potential new or reintroduced sports events (SDEs) for the 2032 Brisbane Olympics:

CRICKET. With its fast-paced gameplay and significant commercial potential, the introduction of cricket would greatly enhance the attractiveness and economic benefits of the Olympics.

MARTIAL ARTS. Martial arts enjoys widespread attention globally, is gender-inclusive, and has relatively low costs. It can enrich the cultural diversity of the Olympics.

SQUASH. Squash attracts young audiences with its intense competition. Its small venue requirements and low-cost characteristics also align with the concept of sustainable development.

In addition, we have recommended the following potential sports that align with development trends for the 2036 Olympics:

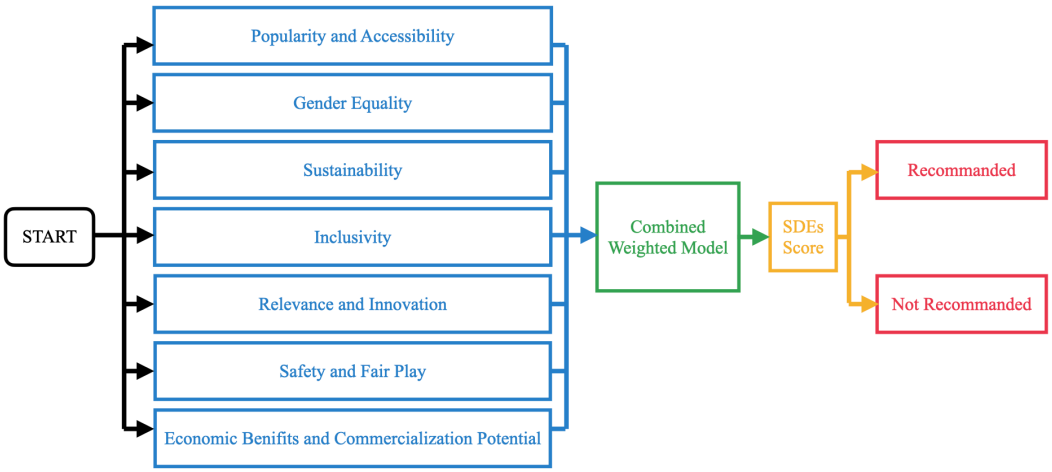
ELECTRONIC SPORTS. Electronic sports, with their well-established tournament facilities and a broad audience base, demonstrate immense economic value and attractiveness.

ULTIMATE FRISBEE. Ultimate Frisbee, with its environmentally friendly characteristics and low costs, perfectly aligns with the Olympics' pursuit of sustainable sports events.

BEACH RUGBY. Beach Rugby, with its flexible scheduling, intense competition, and high entertainment value, has become an Olympic event with great potential.

In summary, our evaluation results emphasize the irreplaceable nature of traditional Olympic events within the Games, while also highlighting the importance of moderately introducing innovative events for the further promotion of the Olympics globally. Therefore, we recommend that the International Olympic Committee (IOC), when considering additions or deletions of events for future Olympics, should respect the core status of traditional events while also focusing on attracting younger audiences and diversified viewer groups, in order to achieve the sustained development and prosperity of the Olympics.

Best regards



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APPENDEX: MATLAB CODE FOR THE MODEL

```
%CWM
data_last=jisuan(data_get,zhibiao_label);
[n,~]=size(data_last);
A_data = data_last ./ repmat(sum(data_last.*data_last) .^ 0.5, n, 1)
[score_cengci, quan_cengci] = layer_analysis(panju, A_data);
[score_entropy, quan_entropy] = entropy_method(A_data);
[score_variation, quan_variation] = variation_coefficient(A_data);
W = [quan_cengci, quan_entropy', quan_variation'];
Score_All=[score_cengci, score_entropy, score_variation];
[~,R2] = corr(W,'type','Kendall');
R2(isnan(R2))=0;
R=R2(1,2:end);
W1=W';
if sum(R)<0.05*length(R)
    Wc=sum(W1)/sum(sum(W1));
    disp('Consistency check passed ')
else
    disp('Consistency check failed')
    weights_critic = critic_method(W);
    Wc = weights_critic .* W;
    Wc = sum(Wc, 2)';
end
s=A_data*Wc';
Score=100*s/max(s);
[AAA,BBB]=sort(Score,'descend');
data_resultA(:,1)= data_table1( BBB,1);
data_resultA(:,2)= data_table1( BBB,2);
data_resultA(:,3)= array2table(AAA);

%MCTOPSIS
data_last = jisuan(data_get, zhibiao_label);
```

```
A_data = data_last;
W = 1 / length(zhibiao_label) * ones(1, length(zhibiao_label));
scoreA = TOPSIS(A_data, W);
for n = 1:100 % Randomized weight analysis
    for i = 1:length(W) - 1
        rand1 = rand(1) * 0.5 * max(W);
        rand2 = rand(1);
        if(rand2 > 0.5)
            rand1 = W(i) - rand1;
        else
            rand1 = W(i) + rand1;
        end
        W1(i) = rand1;
    end
    W2 = W1;
    W2(length(W)) = 1 - sum(W1);
    W_all(n, :) = W2;
end
for N = 1:100
    score_all(:, N) = TOPSIS(A_data, W_all(N, :));
end
mean_score_all = mean(score_all, 2);
std_score_all = std(score_all, 0, 2);
[AAA, BBB] = sort(mean_score_all, 'descend');
data_resultA(:, 1) = data_table1(BBB, 2);
data_resultA(:, 2) = array2table(AAA);
data_resultA(:, 3) = num2cell
```